

Differential Flatness–Locked Control

From Geometric Reduction to DARE

1 Setup: Gravity Breaks $\text{SO}(3)$

A rigid body in $\text{SE}(3)$ has state $(\mathbf{p}, \dot{\mathbf{p}}, R, \boldsymbol{\omega}) \in \mathbb{R}^3 \times \mathbb{R}^3 \times \text{SO}(3) \times \mathbb{R}^3$ —twelve dimensions. The translational dynamics are

$$m\ddot{\mathbf{p}} = R\mathbf{f}_B - mg\mathbf{e}_3 \quad (1)$$

where $\mathbf{f}_B$ is the body-frame force and $\mathbf{e}_3 = (0, 0, 1)^\top$.

For a quadrotor with collective thrust T along the body z -axis, $\mathbf{f}_B = T\mathbf{e}_3$. The key geometric consequence: the *required* thrust direction is

$$R\mathbf{e}_3 = \frac{\ddot{\mathbf{p}} + g\mathbf{e}_3}{\|\ddot{\mathbf{p}} + g\mathbf{e}_3\|} \quad (2)$$

This is a point on S^2 —it **locks** two of the three $\text{SO}(3)$ degrees of freedom (roll, pitch). The surviving freedom is the $\text{SO}(2)$ fiber: yaw ψ .

In the quaternion picture $q = (w, x, y, z) \in S^3$:

$$\underbrace{S^3}_{\text{dim}=3} \xrightarrow{w \text{ killed by } \|q\|=1} \underbrace{\mathfrak{so}(3) \cong \mathbb{R}^3}_{\text{dim}=3} \xrightarrow{g \text{ locks roll, pitch}} \underbrace{\mathbb{R}^1 \cong \text{yaw}}_{\text{dim}=1}$$

2 Flat Output and the Flat Map

$$\boxed{\mathbf{y} = (p_x, p_y, p_z, \psi) \in \mathbb{R}^3 \times S^1} \quad (3)$$

Claim (Differential Flatness). Every state and input is an algebraic function of $\mathbf{y}$ and its derivatives up to order four.

The flat map $\Phi : (\mathbf{y}, \dot{\mathbf{y}}, \ddot{\mathbf{y}}, \mathbf{y}^{(3)}, \mathbf{y}^{(4)}) \mapsto (\mathbf{x}, \mathbf{u})$ proceeds in four steps.

Step 1: Thrust magnitude. Define the *total acceleration vector*

$$\mathbf{t} = \ddot{\mathbf{p}} + g\mathbf{e}_3 \quad (4)$$

Then the collective thrust is

$$T = m \|\mathbf{t}\| \quad (5)$$

Step 2: Rotation matrix. The body z -axis is $\mathbf{z}_B = \mathbf{t}/\|\mathbf{t}\|$. Given yaw angle ψ , define an intermediate vector

$$\mathbf{x}_C = (\cos \psi, \sin \psi, 0)^\top$$

Then the body axes are

$$\mathbf{y}_B = \frac{\mathbf{z}_B \times \mathbf{x}_C}{\|\mathbf{z}_B \times \mathbf{x}_C\|}, \quad \mathbf{x}_B = \mathbf{y}_B \times \mathbf{z}_B, \quad R = [\mathbf{x}_B \quad \mathbf{y}_B \quad \mathbf{z}_B] \quad (6)$$

This is well-defined whenever $\mathbf{z}_B$ is not parallel to $\mathbf{x}_C$ (i.e. the quadrotor is not exactly inverted along the yaw heading).

Step 3: Angular velocity. From $\dot{R} = R[\boldsymbol{\omega}]_\times$ and the chain rule applied to (6):

$$\boldsymbol{\omega} = \begin{pmatrix} \mathbf{y}_B^\top \dot{\mathbf{z}}_B \\ -\mathbf{x}_B^\top \dot{\mathbf{z}}_B \\ \dot{\psi} (\mathbf{e}_3^\top \mathbf{z}_B) \end{pmatrix} \quad (7)$$

where $\dot{\mathbf{z}}_B = \frac{d}{dt}(\mathbf{t}/\|\mathbf{t}\|)$ depends on $\mathbf{p}^{(3)}$.

Step 4: Torques. From Euler's equation $\boldsymbol{\tau} = J\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times J\boldsymbol{\omega}$, which requires $\mathbf{p}^{(4)}$ and $\ddot{\psi}$.

Collecting everything:

State: $\mathbf{x} = (\mathbf{p}, \dot{\mathbf{p}}, R(\ddot{\mathbf{p}}, \psi), \boldsymbol{\omega}(\mathbf{p}^{(3)}, \dot{\psi}))$

Input: $\mathbf{u} = (T(\ddot{\mathbf{p}}), \boldsymbol{\tau}(\mathbf{p}^{(4)}, \ddot{\psi}))$

(8)

3 Control Architecture

The flat map converts the problem into two layers:

3.1 Outer Loop: Trajectory in Flat Space

Plan or optimize a reference trajectory $\mathbf{y}^*(t)$ in $\mathbb{R}^3 \times S^1$. Any trajectory planner works here—polynomials, B-splines, or sampling-based MPC (e.g. MPPI). The key point: *all planning happens in flat space*. No $\text{SO}(3)$ constraints.

Compute the reference state and input via the flat map:

$$\mathbf{x}^*(t) = \Phi_x(\mathbf{y}^*(t), \dots, \mathbf{y}^{*(4)}(t)), \quad \mathbf{u}^*(t) = \Phi_u(\mathbf{y}^*(t), \dots, \mathbf{y}^{*(4)}(t))$$

3.2 Inner Loop: DARE-Based Tracking

Linearize the full nonlinear dynamics $\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u})$ around $(\mathbf{x}^*, \mathbf{u}^*)$:

$$\delta \dot{\mathbf{x}} = A(t) \delta \mathbf{x} + B(t) \delta \mathbf{u}, \quad A = \left. \frac{\partial f}{\partial \mathbf{x}} \right|_*, \quad B = \left. \frac{\partial f}{\partial \mathbf{u}} \right|_* \quad (9)$$

Discretize at time step Δt (e.g. via ZOH) to obtain (A_d, B_d) . The tracking gain is obtained by solving the **Discrete Algebraic Riccati Equation**:

$$P = A_d^\top P A_d - A_d^\top P B_d (R_{\text{lqr}} + B_d^\top P B_d)^{-1} B_d^\top P A_d + Q$$

(10)

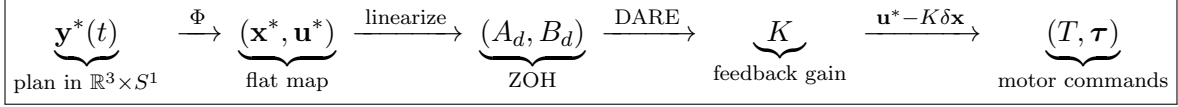
with the feedback gain

$$K = (R_{\text{lqr}} + B_d^\top P B_d)^{-1} B_d^\top P A_d \quad (11)$$

The applied input is:

$$\mathbf{u} = \mathbf{u}^* - K(\mathbf{x} - \mathbf{x}^*) \quad (12)$$

4 Summary: The Complete Pipeline



The entire attitude control problem is *absorbed* by the flat map. Gravity, through equation (2), is not a nuisance—it is the structural reason flatness exists. The DARE (10) is a black-box that converts the linearized tracking problem into a gain matrix; its internals are irrelevant to the geometric story.